



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

Calculus 1 - 104003

Exam B

February 18, 2021

Your ID Number:

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First name and Surname:

Guidelines

1. **Duration: 3 hours.** Use of calculators, personal dictionaries, electronic devices, reference materials, personal notes or any other extra material is not allowed.
2. Explain your solutions, quote theorems you are using.
No credit will be given for non-justified answers!.
Write clear and complete answers for each problem in the same page.

1. (a) Let

$$f(x) = \frac{\sin(ax)}{x} + \frac{\sqrt{ax^2+1}-1}{x}.$$

Find all positive real numbers a for which $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow \infty} f(x)$. (10p)

(b) Compute $\lim_{x \rightarrow \infty} \frac{e^{-x+1} \cos(3x^2-3) + \sqrt{4x}}{x}$. (6p)

Answer

$a > 0 \Rightarrow a \neq 0$

(a)
$$\lim_{x \rightarrow 0} \frac{\sin ax}{x} + \frac{\sqrt{ax^2+1}-1}{x} = \lim_{x \rightarrow 0} a \cdot \frac{\sin ax}{ax} + \frac{(\sqrt{ax^2+1}-1)(\sqrt{ax^2+1}+1)}{x(\sqrt{ax^2+1}+1)} =$$

$$\lim_{x \rightarrow 0} a \cdot \frac{\sin ax}{ax} + \frac{ax^2+1-1}{x(\sqrt{ax^2+1}+1)} = \lim_{x \rightarrow 0} a \cdot \frac{\sin ax}{ax} + \frac{ax}{\sqrt{ax^2+1}+1}.$$

We already know that, since $x \rightarrow 0$ implies $ax \rightarrow 0$, $\frac{\sin ax}{ax} \xrightarrow{x \rightarrow 0} 1$ and $\frac{ax}{\sqrt{ax^2+1}} \xrightarrow{x \rightarrow 0} \frac{0}{2}$. So,

$$\lim_{x \rightarrow 0} a \frac{\sin ax}{ax} + \frac{ax}{\sqrt{ax^2+1}+1} = a.$$

On the other hand,

$$\lim_{x \rightarrow \infty} \frac{\sin ax}{x} + \frac{\sqrt{ax^2+1}-1}{x} = \lim_{x \rightarrow \infty} \frac{\sin ax}{x} + \sqrt{\frac{ax^2+1}{x^2}} - \frac{1}{x} =$$

$$0 + \sqrt{a} - 0 = \sqrt{a}$$

We want thus $a = \sqrt{a} \Leftrightarrow a^2 = a \Leftrightarrow a=0$ or $a=1$, but $a > 0$.

So the only solution is $\boxed{a=1}$.

(b)
$$\lim_{x \rightarrow \infty} \frac{e^{-x+1} \cos(3x^2-3) + \sqrt{4x}}{x} = \lim_{x \rightarrow \infty} \frac{e^{-x+1} \cos(3x^2-3)}{x} + \sqrt{\frac{4x}{x^2}} \rightarrow 0$$

$= 0.$

It is too
since if not
 $\sqrt{4x}$ has no sense

2. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} \frac{x}{\ln x} + e^a x & \text{if } x > 0 \\ \cos x + x - 1 & \text{if } x \leq 0. \end{cases}$$

- (a) Determine for which values of a , the function is continuous at $x = 0$. (7p)
 (b) Determine for which values of a , the function is differentiable at $x = 0$. (8p)
 (c) For all the values found in item (b), describe the tangent line for the curve at $(0, f(0))$. (5p)

Answer

(a) $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{x}{\ln x} + e^a x = 0$.

Let us first compute $\lim_{x \rightarrow 0^+} \frac{x}{\ln x} = 0$.

$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \cos x + x - 1 = 1 + 0 - 1 = 0 = f(0)$.

So f is continuous at $x=0$ for all $a \in \mathbb{R}$.

(b) For f to be differentiable at $x=0$, both $\lim_{h \rightarrow 0^+} \frac{f(h)-f(0)}{h}$ and $\lim_{h \rightarrow 0^-} \frac{f(h)-f(0)}{h}$ must exist, and coincide.

$$\lim_{h \rightarrow 0^+} \frac{f(h)-f(0)}{h} = \lim_{h \rightarrow 0^+} \left(\frac{h}{\ln h} + e^a h - 0 \right) \cdot \frac{1}{h} = \lim_{h \rightarrow 0^+} \left(\frac{1}{\ln h} + e^a \right) = e^a.$$

$$\lim_{h \rightarrow 0^-} \frac{f(h)-f(0)}{h} = \lim_{h \rightarrow 0^-} \left(\cos h + h - 1 \right) \cdot \frac{1}{h} = \lim_{h \rightarrow 0^-} \frac{\cos h - 1}{h} + \frac{h}{h} =$$

$$\lim_{h \rightarrow 0^-} \frac{\cos h - 1}{h} + 1 = 1$$

But $\lim_{h \rightarrow 0^-} \frac{\cos h - 1}{h} \stackrel{\text{L'Hôpital}}{=} \lim_{h \rightarrow 0^-} \frac{-\sin h}{1} = 0$

So, f will be differentiable at $x=0 \Leftrightarrow e^a = 1 \Leftrightarrow \boxed{a=0}$.

(c) For $a=0$, the tangent line at $(0, f(0)) = (0, 0)$ has equation $y = mx + b$, where $m = f'(0) = 1$, so $y = x + b$. Since it passes through $(0, 0)$, we get $0 = 0 + b$, so $b = 0$ and the tangent is $\boxed{y = x}$.

3. Let $f(x) = x^3 + ax^2 + bx + c$.

- (a) Show that the function $f(x)$ can have at most two local extremes. (4p)
- (b) Give an example of a function like this with two local extremes. (4p)
- (c) Give an example of a function like this without extremes. (4p)
- (d) Can you find an example of a function like this with exactly one local extreme? Justify. (6p)

Answer

(a) The domain of f is $\mathbb{R}$, the possible local extremes are the roots of $f'(x) = 3x^2 + 2ax + b$. Since $f'(x)$ is quadratic, it has at most two roots, so, there are at most two local extremes.

(b) ^{for example} Suppose $f'(x) = 3(x-3)(x-1) = 3(x^2 - 4x + 3) = 3x^2 - 12x + 9$,
so $2a = -12$ and $b = 9$, that is $\boxed{a = -6}$ and $\boxed{b = 9}$.

We see that there are no restrictions for the choice of c , we thus choose $\boxed{c = 0}$.

$$f(x) = x^3 - 6x^2 + 9x.$$

(c) We look for $a, b, c \in \mathbb{R}$ such that $f'(x)$ has no roots. For this, its discriminant, that is $(2a)^2 - 12b$ must be negative.

$4a^2 - 12b = 4(a^2 - 3b) < 0 \Leftrightarrow a^2 < 3b$, so, choose for example $a = 0$ and $b = 3$. That is, $f'(x) = 3x^2 + 3 = 3(x^2 + 1)$.

Again, we get no restrictions on c , so we can choose:

$$f(x) = x^3 + 3x.$$

(d) We ask $f'(x)$ to have exactly one root, so we must have:

$$\boxed{a^2 = 3b}, \text{ so } b = \frac{a^2}{3}$$

We get that in this case $x = -\frac{2a}{6} = -\frac{a}{3}$ is the unique root of $f'(x) = 3x^2 + 2ax + \frac{a^2}{3}$.

$f'(x) = 3\left(x + \frac{a}{3}\right)^2$ is positive except for $x = -\frac{a}{3}$,
that is, $f'\left(-\frac{a}{3}\right) = 0$ and $f'(x) > 0 \forall x \neq -\frac{a}{3}$, so $x = -\frac{a}{3}$
cannot be a local extreme.

4. Let

$$f(x) = \begin{cases} x^2 & \text{if } -1 \leq x \leq 1 \\ (x-2)^2 & \text{if } 1 < x \leq 3. \end{cases}$$

- (a) Find its critical points. (4p)
- (b) Find its local extremes (4p)
- (c) Find its global extremes. (4p)

Justify using theory, not using the graph.

Answer

(a) The critical points are the points x in the domain of f such that either $f'(x)=0$, or the extremes of the intervals.

For $x \in (-1, 1)$, $f'(x)=0 \Leftrightarrow 2x=0 \Leftrightarrow \boxed{x=0}$

For $x \in (1, 3)$, $f'(x)=0 \Leftrightarrow 2(x-2)=0 \Leftrightarrow \boxed{x=2}$, so the critical points of $f(x)$ are $\boxed{0, 2, -1, 1, 3}$.

(b) For $x \in (-1, 1)$, $f'(x) > 0 \Leftrightarrow 2x > 0 \Leftrightarrow x > 0 \Rightarrow f$ inc. on $(0, 1)$
 $f'(x) < 0 \Leftrightarrow x < 0 \Rightarrow f$ decreasing on $(-1, 0)$.

For $x \in (1, 3)$, $f'(x) > 0 \Leftrightarrow 2(x-2) > 0 \Leftrightarrow x > 2 \Rightarrow f$ increasing on $(2, 3)$.
 $f'(x) < 0 \Leftrightarrow x < 2 \Rightarrow f$ decreasing on $(1, 2)$.

So $0, 2$ are local minima and $-1, 1, 3$ are local maxima.

(c) $f(-1)=2 = f(1)=f(3) \Rightarrow -1, 1, 3$ are global maxima.

$f(0)=0 = f(2) \Rightarrow 0, 2$ are global minima.

5. (a) Compute $\int x \ln(x^2) dx$. (8p)
- (b) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable increasing function that satisfies $f(1) = 3$ and $f(2) = 4$. Compute the area of the region bounded by the curves $y = f'(x)$, $y = -x \ln(x^2)$, $x = 1$ and $x = 2$. (8p)

Answer

$$(a) \int x \cdot \ln(x^2) dx = \frac{x^2}{2} \cdot \ln(x^2) - \int \frac{2}{x} \cdot \frac{x^2}{2} dx = \frac{x^2}{2} \ln(x^2) - \int x dx$$

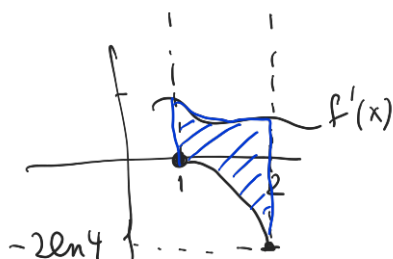
$$v' = x \Rightarrow v = \frac{x^2}{2}$$

$$u = \ln(x^2) \Rightarrow u' = \frac{1}{x^2} \cdot 2x = \frac{2}{x}$$

$$= \boxed{\frac{x^2}{2} \ln(x^2) - \frac{x^2}{2} + C}$$

(b) The only things that we know about $f(x)$ are:

- $f(1) = 3$ and $f(2) = 4$,
- $f(x)$ is differentiable, so $\int f'(x) dx = f(x) + C$
- $f(x)$ is increasing, so $f'(x) > 0$.



$$y = -x \ln(x^2)$$

Remark: for $x \in [1, 2]$, $\ln(x^2) \geq 0$, so $-x \ln(x^2) \leq 0$

We have to compute the area in blue.

$$\int_1^2 f'(x) dx + \int_1^2 (-x \ln x^2) dx = \int_1^2 f'(x) dx + \int_1^2 x \ln x^2 dx =$$

$$f(2) - f(1) + \frac{x^2}{2} (\ln(x^2) - 1) \Big|_1^2 = 4 - 3 + \frac{4}{2} (\ln 4 - 1) - \frac{1}{2} (\ln 1 - 1)$$

$$= \boxed{1 + 2 \ln 4}$$

6. (a) Compute $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$. (6p)
 (b) Determine if $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(2n+1)}}$ is convergent or not. (6p)
 (c) Determine if $\sum_{n=1}^{\infty} \frac{1}{2^n}$ and $\int_1^{\infty} \frac{1}{2^x} dx$ are equal or not. (6p)

Answer

$$(a) \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \frac{1}{n} - \frac{1}{n+1} = \lim_{N \rightarrow \infty} \sum_{n=1}^N \frac{1}{n} - \frac{1}{n+1} =$$

$$\lim_{N \rightarrow \infty} \left(1 - \frac{1}{N+1} \right) = 1.$$

(b) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(2n+1)}}$ is divergent, since all its terms are positive and:

$$\frac{1}{\sqrt{n(2n+1)}} \geq \frac{1}{\sqrt{n(2n+n)}} = \frac{1}{\sqrt{3n^2}} = \frac{1}{\sqrt{3}n} \text{ and } \sum_{n=1}^{\infty} \frac{1}{\sqrt{3}n} \text{ is divergent}$$

$$(c) \sum_{n=1}^{\infty} \frac{1}{2^n} = \sum_{n=1}^{\infty} \left(\frac{1}{2} \right)^n = \frac{\frac{1}{2} - 1}{-\frac{1}{2}} = 1.$$

$$\int_1^{\infty} \frac{1}{2^x} dx = \lim_{n \rightarrow \infty} \int_1^n \frac{1}{2^x} dx = \lim_{n \rightarrow \infty} \frac{1}{\ln \frac{1}{2}} \left(\frac{1}{2} \right)^x \Big|_1^n = \lim_{n \rightarrow \infty} -\frac{1}{\ln 2} \left(\left(\frac{1}{2} \right)^n - \frac{1}{2} \right) =$$

$$-\frac{1}{\ln 2} \left(-\frac{1}{2} \right) = \frac{1}{2 \ln 2}, \text{ so } \boxed{\text{the values are different.}}$$